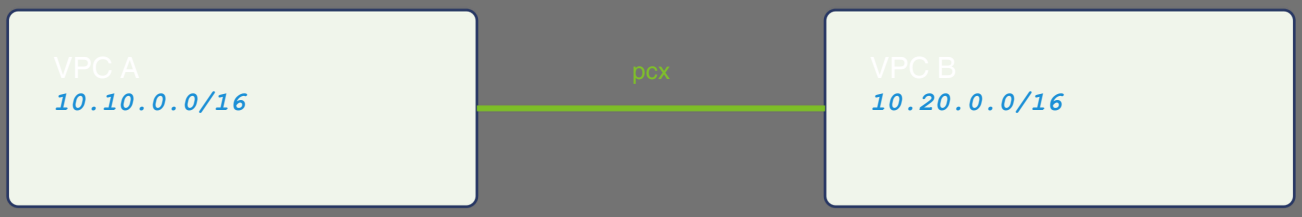
AWS NETWORKING / PROJECT 01

# VPC Peering

## Connecting two VPCs and proving private connectivity

Project report: EC2, custom route tables, security groups and a peering connection




Submited by :-
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# 1. Objective

The goal of this project was to connect two separate AWS Virtual Private Clouds (VPCs) with a VPC peering connection so that workloads in each VPC can talk to each other over private IPv4 addresses, without using the public internet. The build also had to show that each route and security rule has a purpose: non-overlapping CIDR ranges, explicit route table associations, forward and return routes, and tightly scoped inbound rules.

What this report demonstrates:

- Planning non-overlapping VPC and subnet CIDR ranges.
- Creating public and private subnets with explicit route table associations.
- Requesting, accepting and activating a peering connection, then adding routes on both sides.
- Launching EC2 instances in the two VPCs and verifying private connectivity.

# 2. Architecture and address plan

VPC A (10.10.0.0/16) holds a public and a private subnet and is the only VPC with an internet gateway. VPC B (10.20.0.0/16) has a single private subnet and no internet gateway. A peering connection links the two VPCs, so the private subnets reach each other directly using private IP addresses.

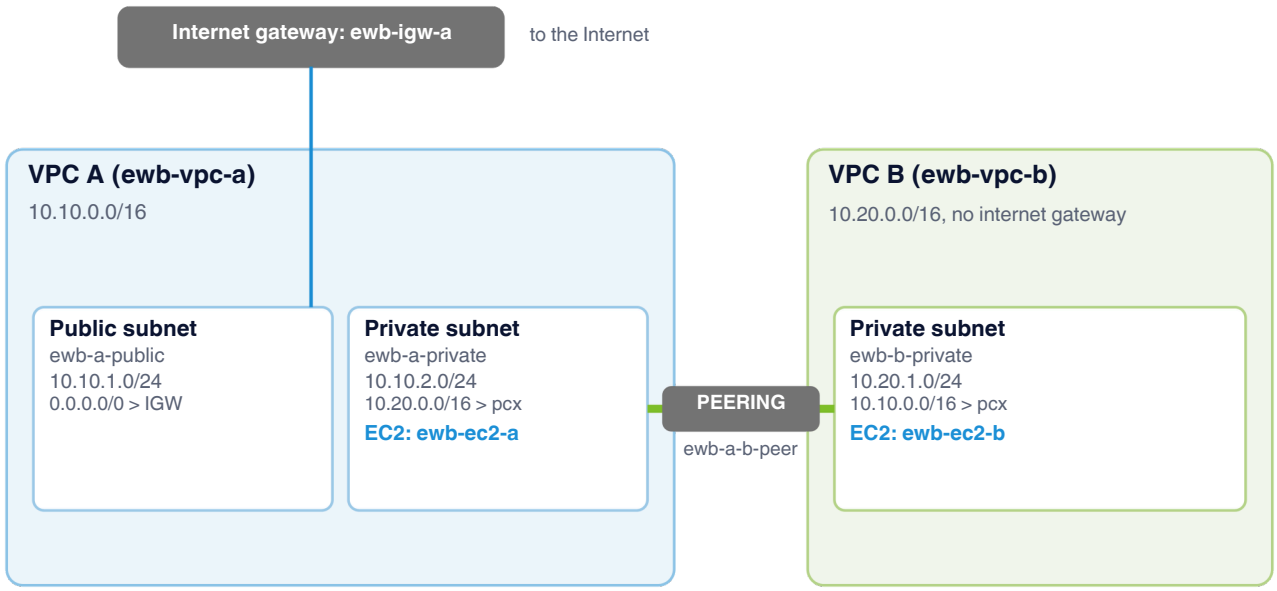



**Figure 1.** Architecture of the project (conceptual). Peering does not give VPC B internet access through VPC A.

## Address plan

| Resource (Name tag) | Placement | IPv4 CIDR |
|---|---|---|
| ewb-vpc-a | VPC A | 10.10.0.0/16 |
| ewb-a-public | VPC A, public subnet | 10.10.1.0/24 |
| ewb-a-private | VPC A, private subnet | 10.10.2.0/24 |
| ewb-vpc-b | VPC B | 10.20.0.0/16 |
| ewb-b-private | VPC B, private subnet | 10.20.1.0/24 |

The two VPC ranges do not overlap, which is a hard requirement: AWS refuses to create a peering connection between VPCs with overlapping CIDRs. Every /24 subnet sits inside its own VPC range.

# 3. Resource inventory

The table lists the resources created in the **Asia Pacific (Sydney) ap-southeast-2** Region, with the IDs shown in the AWS console.

| Resource | Name ewb-vpc- | ID / detail |
|---|---|---|
| VPC A | a ewb-vpc-b | vpc-0e13b1275c4404f86 (10.10.0.0/16) |
| VPC B | ewb-a-public | vpc-0b570eaf09596cbfa (10.20.0.0/16) |
| Subnet | ewb-a-private | subnet-093184ce1cf20194c (10.10.1.0/24) |
| Subnet | ewb-b-private | subnet-0c3c48d885ebe... (10.10.2.0/24) |
| Subnet | ewb-igw-a ewb- | subnet-09aac6debd58277e4 (10.20.1.0/24) |
| Internet gateway | rt-a-public ewb- | igw-083541cabc1ddaac6, attached to ewb-vpc-a |
| Route table | rt-a-private | rtb-0f1d18e9fea198201, associated with ewb-a-public |
| Route table | ewb-rt-b-private | Created in VPC A for ewb-a-private |
| Route table Peering | ewb-a-b-peer | rtb-0a50e327a9e62155e, associated with ewb-b-private |
| connection EC2 | ewb-ec2-a ewb- | pcx-0ee8e4be0ede9ba99, status Active |
| instance | ec2-b | i-0b2fa2f3b7306a262, t3.micro, ap-southeast-2a |
| EC2 instance | | i-08a657c4b1065e30b, t3.micro, ap-southeast-2a |

# 4. Implementation

## Step 1: Create VPC A

In the VPC console I chose Create VPC with **VPC only**, so that every subnet and route could be controlled manually. The VPC was named ewb-vpc-a with the IPv4 CIDR 10.10.0.0/16, no IPv6 CIDR and default tenancy. The VPC reached the Available state.

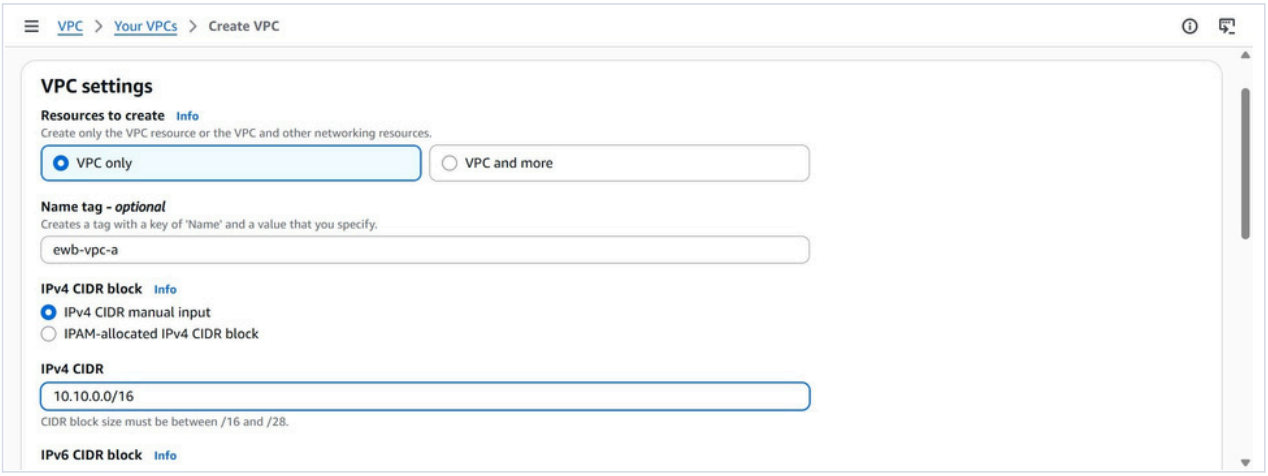



**Figure 2.** Create VPC wizard: VPC only, name ewb-vpc-a, IPv4 CIDR 10.10.0.0/16.

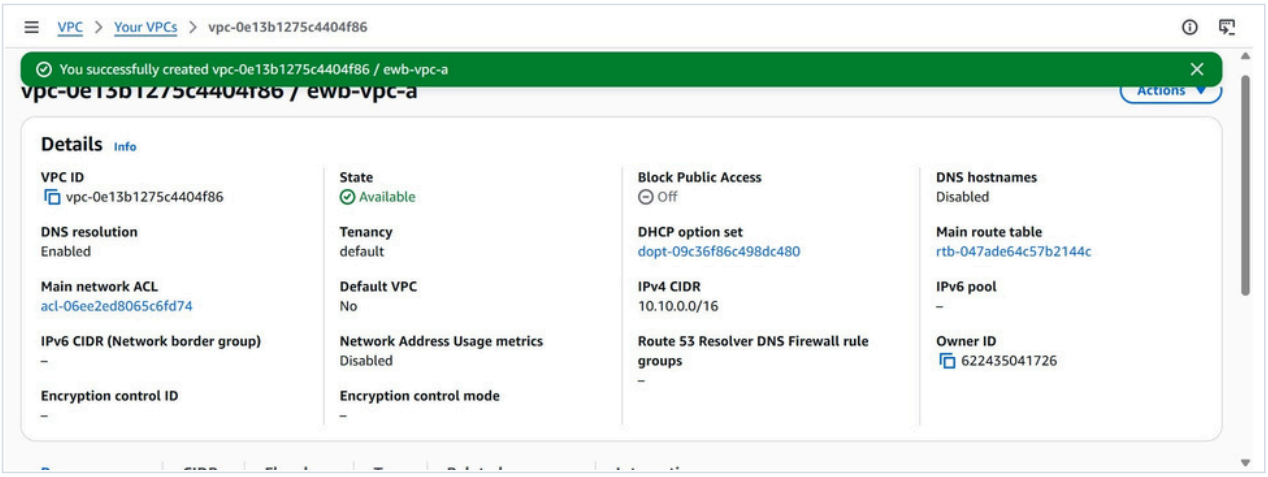



**Figure 3.** VPC ewb-vpc-a (vpc-0e13b1275c4404f86) created successfully and in the Available state.

## Step 2: Create the subnets in VPC A

Two subnets werecreatedinVPC A:ewb-a-public (10.10.1.0/24) and `ewb-a-private` (10.10.2.0/24). The subnet associations screen in Figure 7 lists both subnets with their CIDRs.

## Step 3: Attach the internet gateway

Aninternet gateway named ewb-igw-awas created and attached to ewb-vpc-a. An internet gateway belongs to the VPC and is used as a route target in the route table, so attaching it alone does not give a subnet internet access.

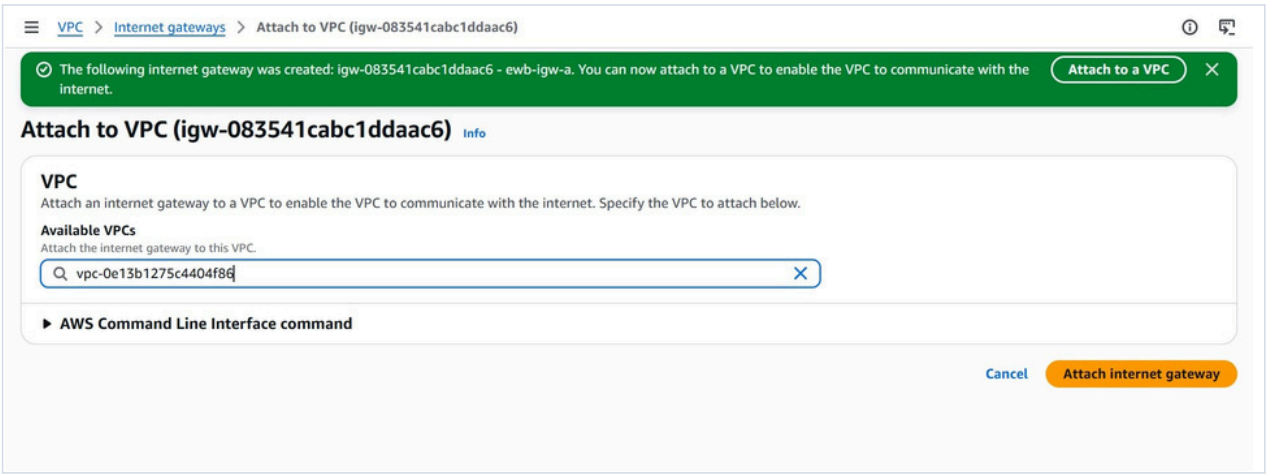



**Figure 4.** Attaching internet gateway ewb-igw-a (igw-083541cabc1ddaac6) to vpc-0e13b1275c4404f86.

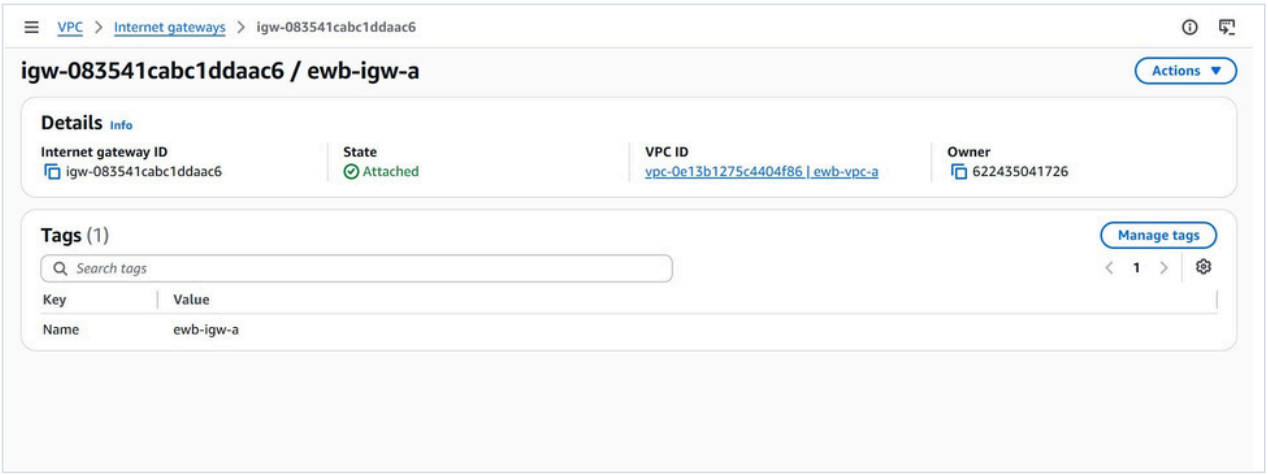



**Figure 5.** Internet gateway ewb-igw-a is in the Attached state, linked to ewb-vpc-a.

## Step 4: Create the route tables in VPC A

Acustom route table ewb-rt-a-publicwas created for VPC A, and ewb-a-public was explicitly associated with it. A second route table, ewb-rt-a-private, was created for the private subnet. It keeps only the local route and has no internet default route, so the private subnet stays private.

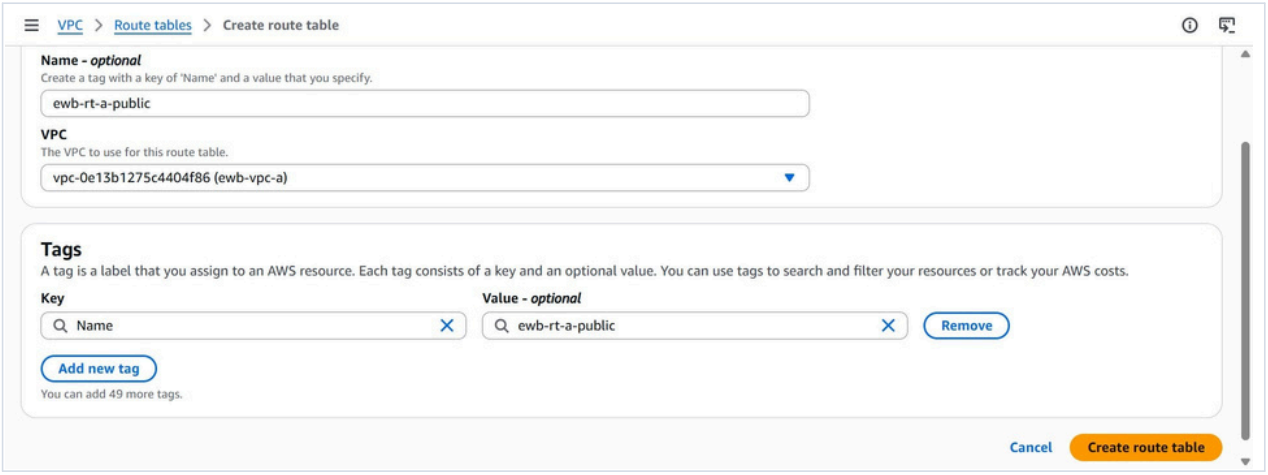



**Figure 6.** Creating route table ewb-rt-a-public in vpc-0e13b1275c4404f86 (ewb-vpc-a).

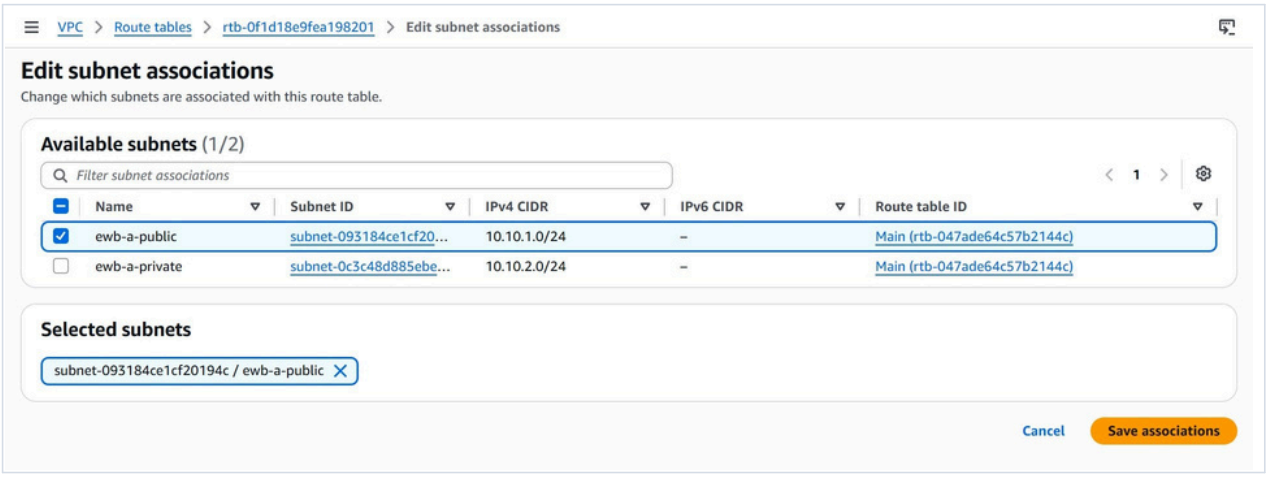



**Figure 7.** Explicit subnet association: ewb-a-public (10.10.1.0/24) selected for ewb-rt-a-public.

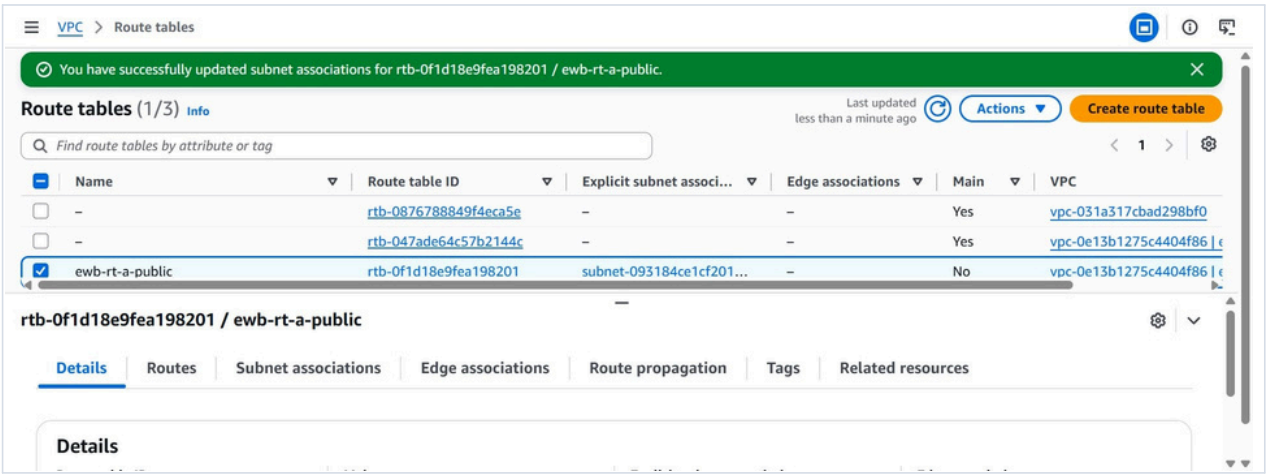



**Figure 8.** Subnet associations updated successfully; ewb-rt-a-public is now associated with subnet-093184ce1cf20...

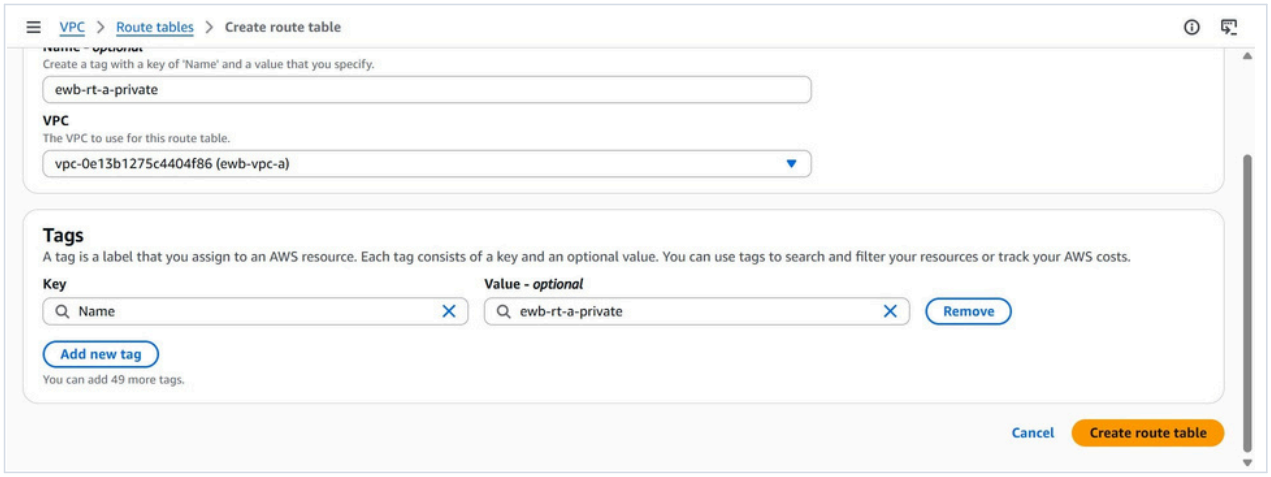



**Figure 9.** Creating route table ewb-rt-a-private in ewb-vpc-a.

## Step 5: Create VPC B, its subnet and route table

VPCB(ewb-vpc-b, 10.20.0.0/16) was created with nointernet gateway. Inside it, the private subnet ewb-b-private (10.20.1.0/24) and the route table ewb-rt-b-private were created, and the subnet was explicitly associated with the route table.

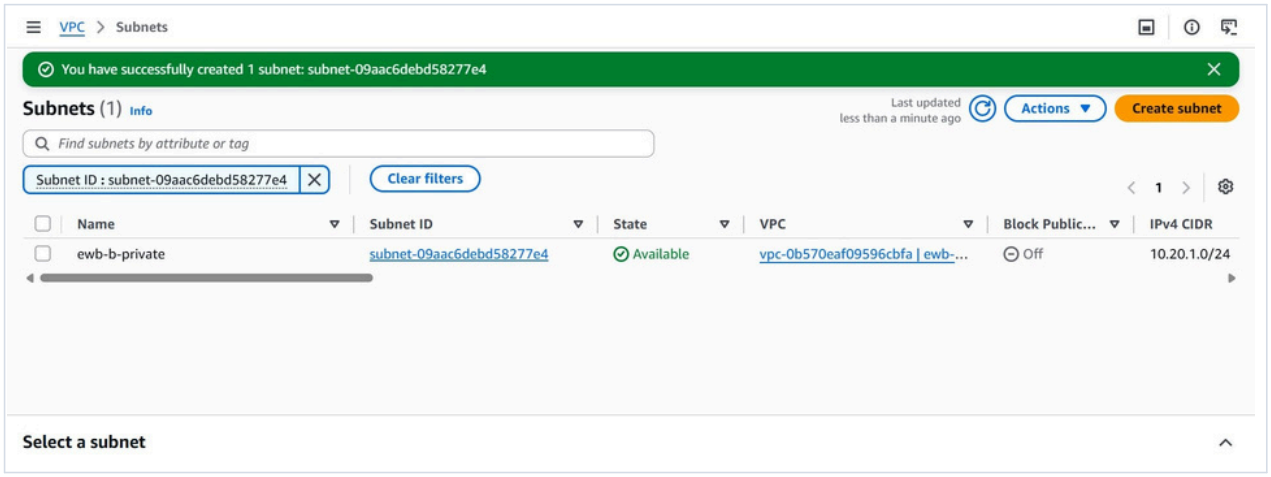



**Figure 10.** Subnet ewb-b-private (subnet-09aac6debd58277e4, 10.20.1.0/24) created and Available in VPC B.

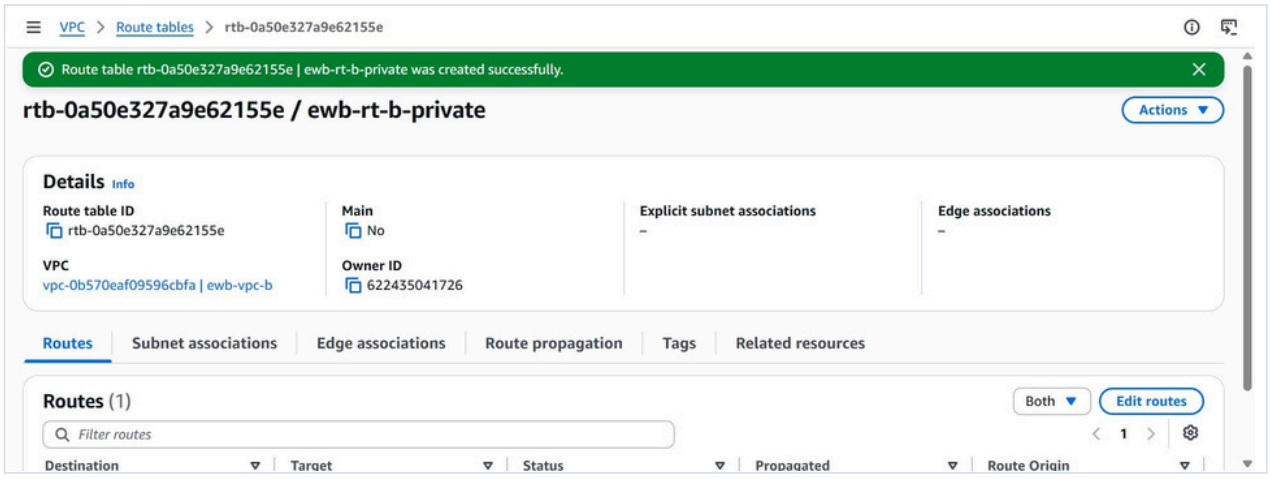



**Figure 11.** Route table ewb-rt-b-private (rtb-0a50e327a9e62155e) created in ewb-vpc-b.

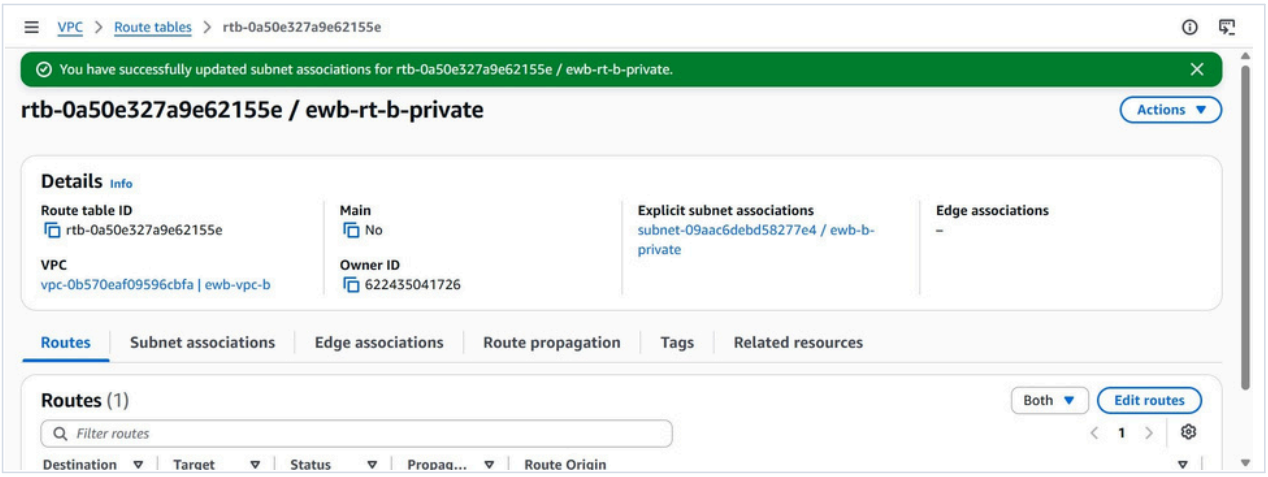



**Figure 12.** Subnet association updated: subnet-09aac6debd58277e4 / ewb-b-private is now explicitly associated with ewb-rt-b-private.

## Step 6: Request and accept the peering connection

A peeringconnection named ewb-a-b-peerwas requestedwith ewb-vpc-a as the requester and ewb-vpc-b as the accepter. Both VPCs are in the same account and Region. The request first showed Pending acceptance. Even within one account, the request must be accepted manually. After acceptance the status changed to **Active**, and the requester CIDR (10.10.0.0/16) and accepter CIDR (10.20.0.0/16) were both shown.

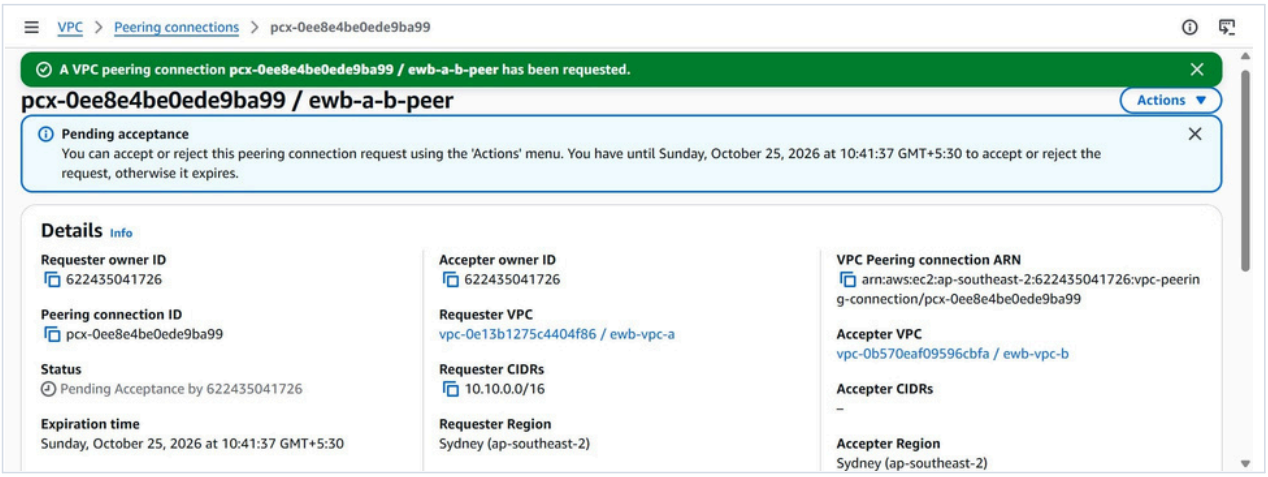



**Figure 13.** Peering connection pcx-0ee8e4be0ede9ba99 (ewb-a-b-peer) requested: status Pending acceptance.

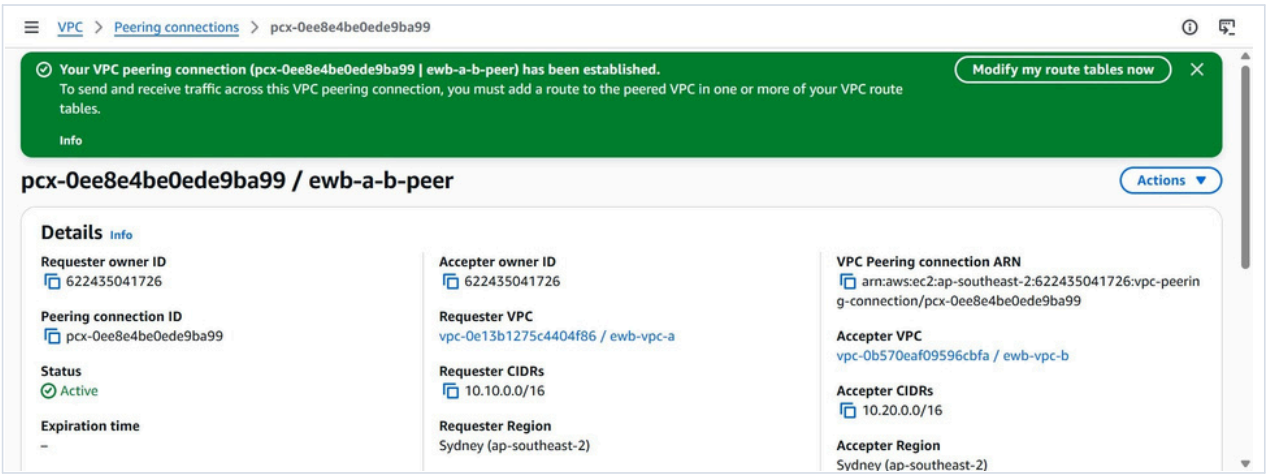



**Figure 14.** Peering connection ewb-a-b-peer established and Active, with requester 10.10.0.0/16 and accepter 10.20.0.0/16.

> **Important:** an Active peering connection does not by itself let traffic flow. It only creates the link. Routes in both VPCs and the security group rules decide whether traffic can use it.

## Step 7: Add routes on both sides

Each sideneeds a routetothe otherVPC, because a reply needs a way back as well as a way out. The routes below were configured, with pcx-0ee8e4be0ede9ba99 as the target for the peer CIDR.

| Route table | Destination | Target | Purpose |
|---|---|---|---|
| ewb-rt-a-private | 10.10.0.0/16 | local | VPC A internal traffic |
| ewb-rt-a-private | 10.20.0.0/16 | pcx (ewb-a-b-peer) | Forward path from VPC A to VPC B |
| ewb-rt-b-private | 10.20.0.0/16 | local | VPC B internal traffic |
| ewb-rt-b-private | 10.10.0.0/16 | pcx (ewb-a-b-peer) | Return path from VPC B to VPC A |
| ewb-rt-a-public | 10.10.0.0/16 | local | VPC A internal traffic |
| ewb-rt-a-public | 0.0.0.0/0 | igw (ewb-igw-a) | Internet access for the public subnet |
| ewb-rt-a-public | 10.20.0.0/16 | pcx (ewb-a-b-peer) | Public subnet to VPC B |

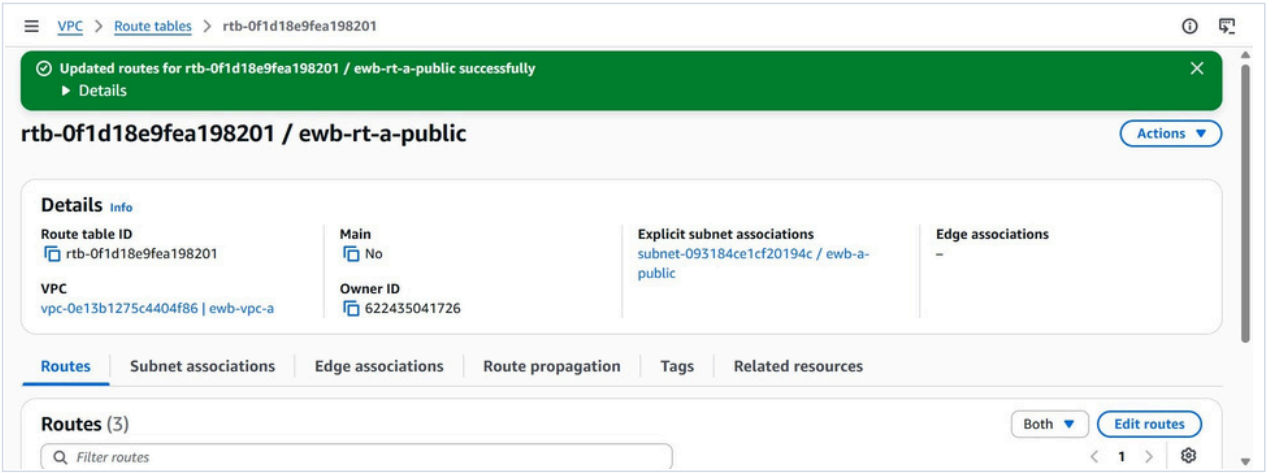



**Figure 15.** Routes updated successfully for ewb-rt-a-public (rtb-0f1d18e9fea198201); the route table now has 3 routes.

The more specific 10.20.0.0/16 route is chosen for traffic to VPC B instead of the broader internet default route, because AWS always picks the most specific matching route.

## Step 8: Launch the EC2 instances

Two `t3.micro` instanceswerelaunched in Availability Zone ap-southeast-2a: ewb-ec2-a for VPC A and `ewb-ec2-b` for VPC B. Both reached the Running state and passed all 3 of 3 status checks.

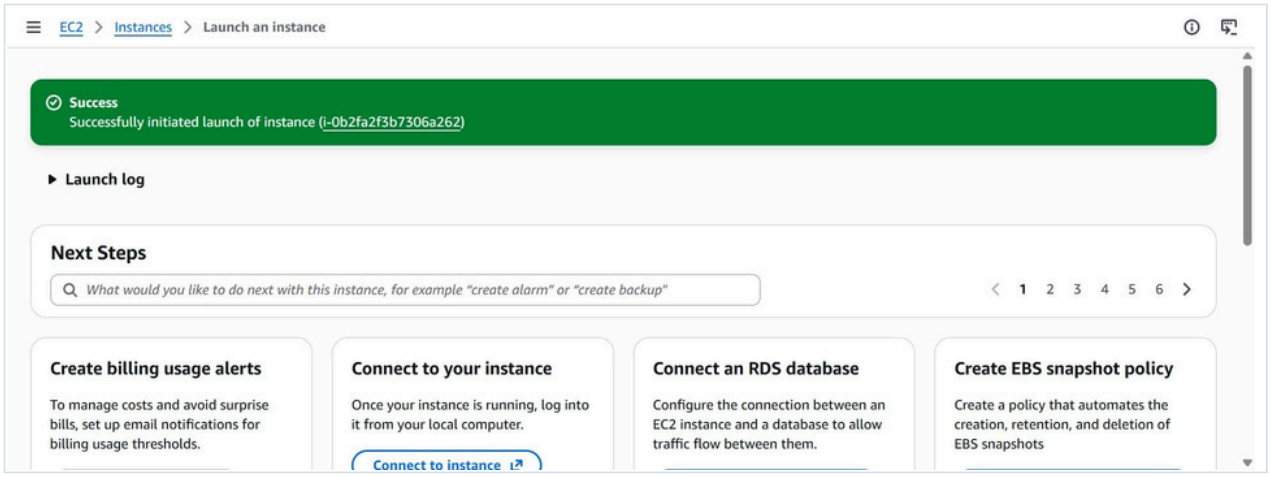



**Figure 16.** EC2 launch initiated successfully for instance i-0b2fa2f3b7306a262.

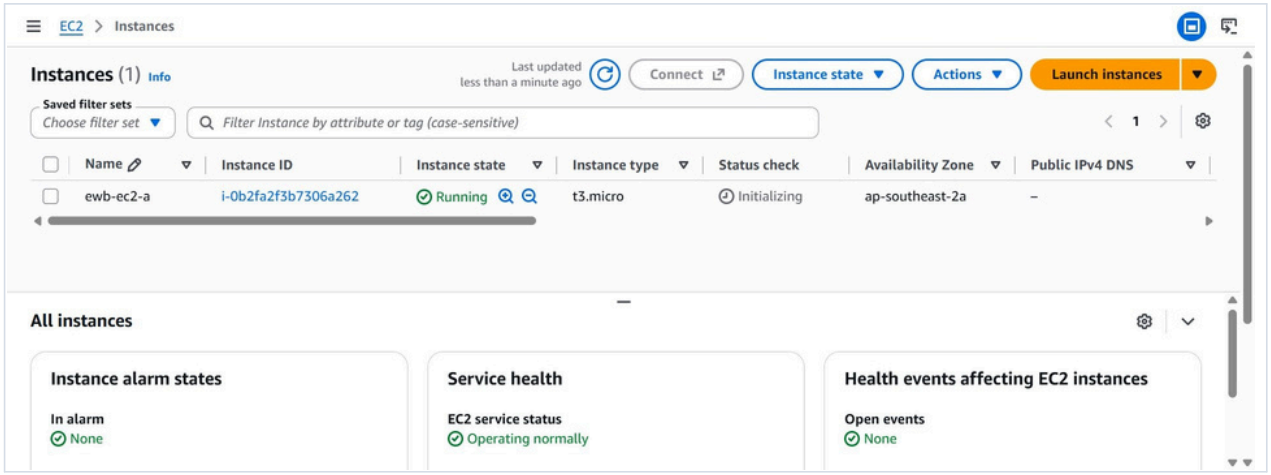



**Figure 17.** ewb-ec2-a (i-0b2fa2f3b7306a262) running; status checks initializing.

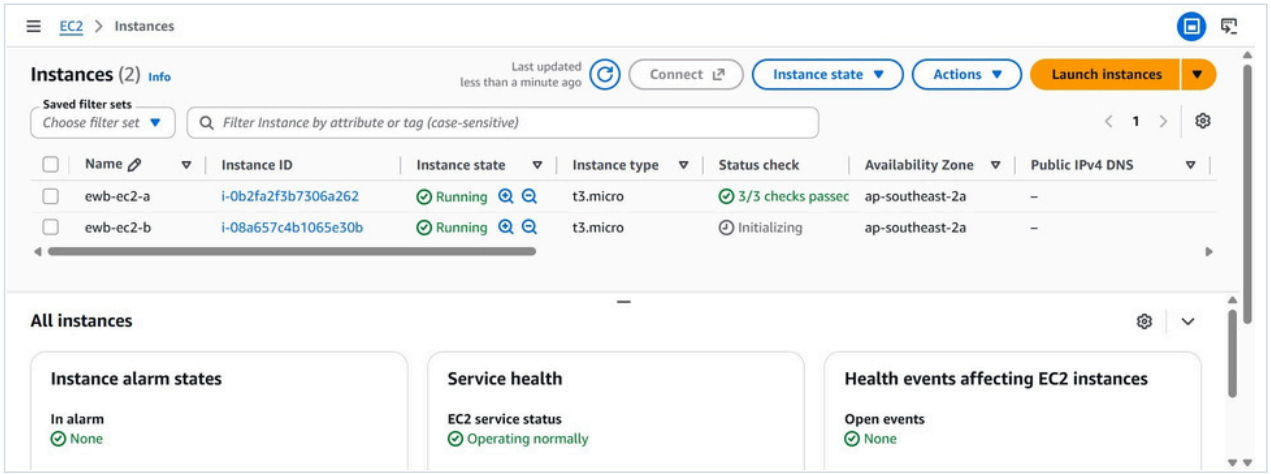



**Figure 18.** Second instance ewb-ec2-b (i-08a657c4b1065e30b) launched; ewb-ec2-a now shows 3/3 checks passed.

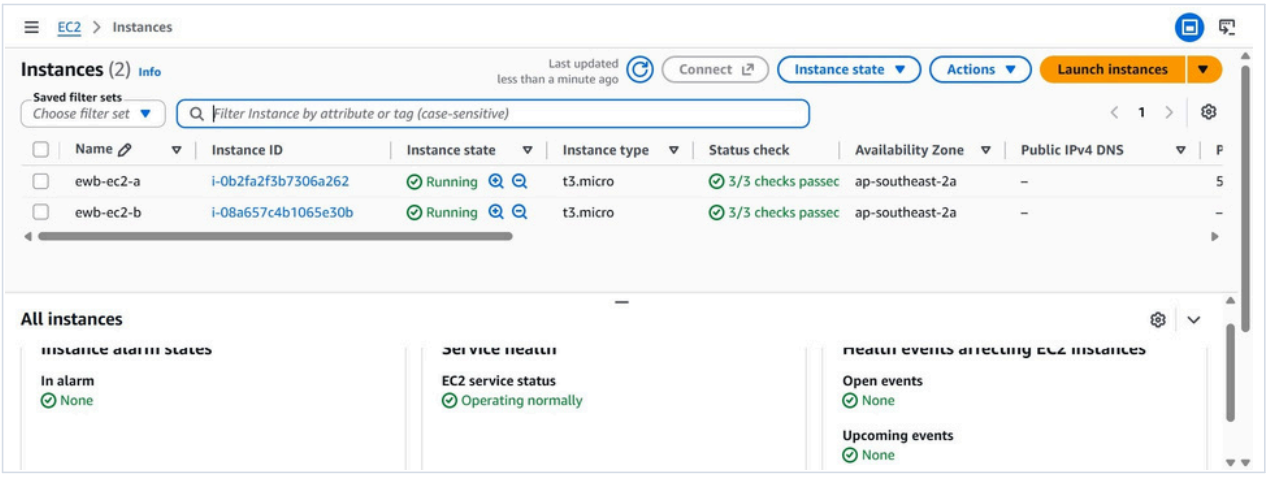



**Figure 19.** Both instances Running with 3/3 status checks passed, ready for connectivity testing.

# 5. Security design

Securitygroups actasstatefulfirewalls, so replies to permitted traffic are allowed automatically. The inbound rules follow the principle of least privilege: only the specific lab hosts are allowed, and nothing is opened to `0.0.0.0/0.`

| Security group ewb- | Type / port | Source |
|---|---|---|
| sg-app-a (VPC A) ewb- | Custom ICMP IPv4, Echo request | Private IP of the instance in VPC B (/32) |
| sg-app-a (VPC A) ewb- | SSH / TCP 22 | Private IP of the administration host (/32) |
| sg-app-b (VPC B) ewb- | Custom ICMP IPv4, Echo request | Private IP of the instance in VPC A (/32) |
| sg-app-b (VPC B) | SSH / TCP 22 | Private IP of the administration host (/32) |

Both application instances are private: they have no public IPv4 address, and the subnets are not routed to an internet gateway except the public subnet in VPC A. The key pair stays on my own computer and is never copied to any instance.

# 6. Testing and verification

Peeringisconsideredsuccessful onlyif traffic actually flows, so the tests below check both directions and also include a controlled failure. All tests use private destination IP addresses.

| Test | Where / command | Expected result |
|---|---|---|
| 01 Baseline | ewb-ec2-a, before peering routes: `ping -c 4 <private IP of ewb-ec2-b>` | No replies |
| 02 Forward path | ewb-ec2-a, after routes: `ping -c 4 <private IP of ewb-ec2-b>`<br>ewb-ec2-b, after routes:<br>SSH to ewb-ec2-b through the bastion path | 4 replies, 0% packet loss |
| 03 Reverse path | EC2 instance details of both private hosts<br>`ping -c 4 <private IP of` | 4 replies, 0% packet loss |
| 04 Private SSH | | Shell on ewb-ec2-b |
| 05 Isolation | Remove only the ICMP Echo rule on ewb-sg-app-b, ping again, then restore the exact rule | No public IPv4 address |
| 06 Controlled failure | | Replies stop, then resume |

Explanation of the controlled failure: removing the Echo request rule changes only the security layer, while the route and peering stay in place. The ping fails, which proves that security groups are evaluated on top of

routing. Restoring the rule brings replies back, so the fault is isolated to that one rule.

# 7. Forward path and return path

**Forward path.** A packet from the instance in VPC A to a 10.20.x.x address matches the 10.20.0.0/16 route in ewb-rt-a-private, whose target is the peering connection. It crosses the peering link and arrives in VPC B, where the destination security group must allow the traffic.

**Return path.** The reply to a 10.10.x.x address matches the 10.10.0.0/16 route in ewb-rt-b-private, which also points at the peering connection. Without that route the reply would have nowhere to go, so a one-way route is not enough. The security group is stateful, so the reply needs no extra inbound rule.

# 8. Key concepts and questions answered

- **Why is the return route needed?** Routing is per direction. If only VPC A knows the way to VPC B, requests arrive but replies cannot get back.
- **Why does an Active peering connection not guarantee a successful ping?** Active only means the link exists. Route tables, security groups and network ACLs still decide whether traffic is allowed.
- **Can VPC B use the internet gateway of VPC A over peering?** No. Peering is not transitive and does not extend internet access, so VPC B stays private.
- **Why must the CIDR ranges not overlap?** Overlapping ranges would make routing ambiguous, so AWS blocks the peering connection.
- **Is peering transitive?** No. A-B plus A-C does not give B-C connectivity.

# 9. Troubleshooting checklist

| Symptom | What tocheck |
|---|---|
| Ping fails after acceptance | Both private route tables, the correct pcx ID and Active state, the Echo request rule on the destination, and the actual host IPs. |
| Peering creation fails | Check every CIDR of both VPCs for overlap, including secondary CIDRs. |
| SSH times out | Instance state, destination IP, subnet route table, then the security group rule, in that order. |
| Permission denied (publickey) | Correct key pair, local identity file and the ec2-user name. Routing is working if SSH authentication was reached. |

# 10. Cleanup plan

To avoid charges, all resources created for this lab should be removed once the evidence is saved:

- Terminate the EC2 instances and confirm their volumes are deleted. Release any Elastic IP and remove the unused key pair.
- Delete the peer routes and the peering connection ewb-a-b-peer.
- Delete the lab security groups, then the subnets and custom route tables.
- Detach and delete ewb-igw-a, then delete both VPCs.
- Verify with the project tag that nothing chargeable remains.

# 11. What I learned

This project showed me that a working network is the combination of several small pieces, and that each piece has one clear job. A VPC peering connection only creates the link. Routes on both sides send the traffic across it, and security groups decide which hosts may talk. I also learned why careful CIDR planning at the start matters, since overlapping ranges cannot be fixed later without rebuilding. Testing from both directions, and breaking one rule on purpose, made it much easier to understand where a connectivity problem can come from. The same ideas apply to many real AWS designs, such as connecting application and database environments privately.

Report prepared by **Methrick Jose Raja**.